



DISCORD ROBOT

鄭聖輝 11303022A

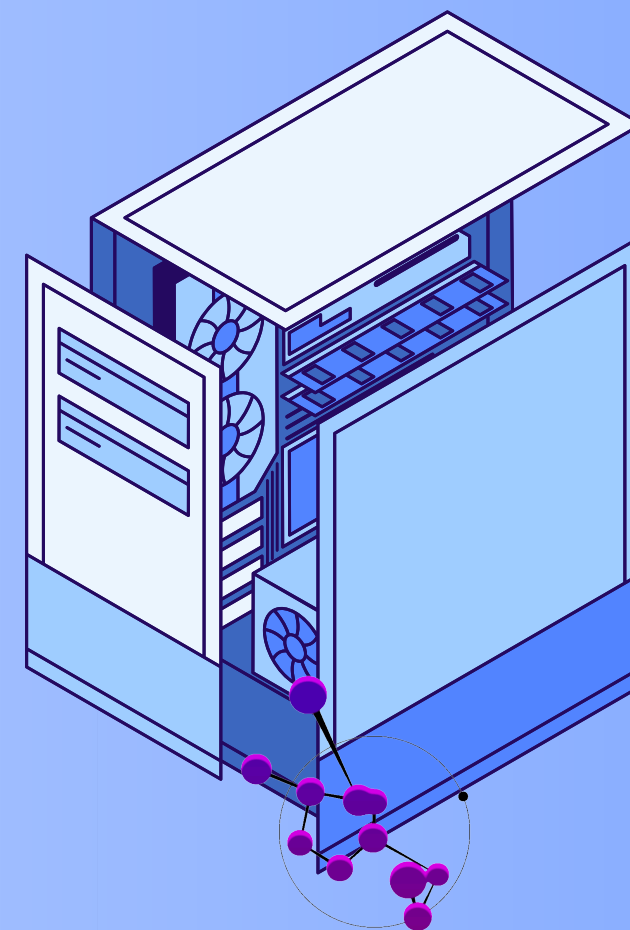
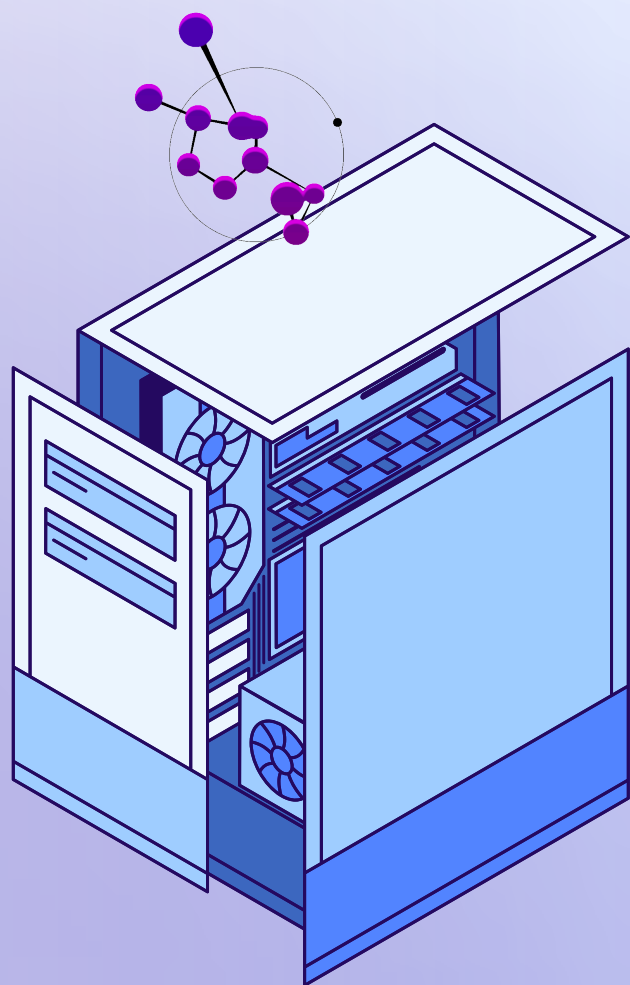
鍾峻亦 11303014A





主題說明

能夠自動化地幫助伺服器管理員執行繁忙瑣碎的任務，例如：歡迎新成員、自動分發身分組、點播音樂、管理伺服器秩序等等功能，最重要的是每位Discord 使用者都能免費打造屬於自己獨一無二的機器人。





KEY

製作動機





製作動機



在Discord裡，有著各式各樣不同的機器人，但每隻機器人的功能都不大相同，導致每次要用不同的功能，就需要再邀請新的機器人進入伺服器，因此希望透過製作機器人，自動化管理工作、增強互動與娛樂，聚集自己所需功能於同一機器人上。





KEY

使用工具





使用工具

1. chat on (debug、製作機器人所需程式碼)
2. colab (將程式碼寫入製作機器人)
3. 11303022A (人工debug)





KEY

製作過程

by chat on





利用chat on生成機器人程式碼並且為每個功能增加註解以方便閱讀。此程式碼中包含General, Fun, Music, Game, Poll, User, Moderation等命令，在Discord聊天室中輸入!help_command則會顯示所以命令所對應的功能。



以下是更新過的 help_command 指令，包含更多的功能描述，讓用戶可以清楚了解可以使用的所有命令：

Code

```
@bot.command()
async def help_command(ctx):
    help_message = """
    Here are the commands you can use:

    **General Commands:**
    !hello - Greets you.
    !help_command - Shows this help message.

    **Fun Commands:**
    !joke - Tells a random joke.
    !rps <rock/paper/scissors> - Play rock, paper, scissors against the bot.

    **Music Commands:**
    !join - Makes the bot join your voice channel.
    !play <url> - Plays music from the given YouTube URL.
    !leave - Makes the bot leave the voice channel.

    **Game Commands:**
    !start_game - Starts a number guessing game (0-100).
    !end_game - Ends the current number guessing game.

    **Poll Commands:**
    !poll <question> <option1> <option2> [option3] ... - Creates a poll.

    **User Commands:**
    !userinfo [@user] - Displays information about a user.

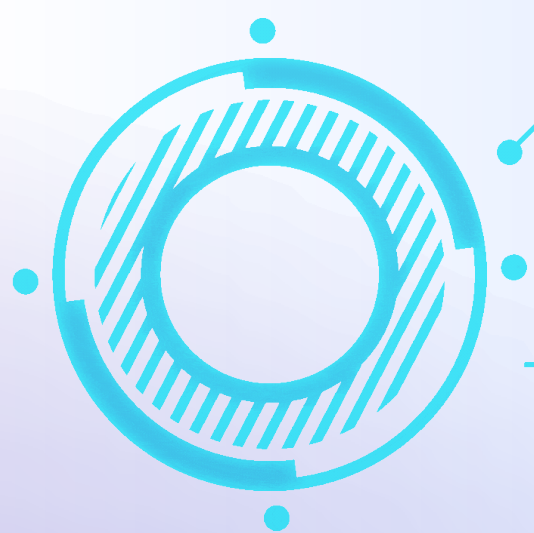
    **Moderation Commands:**
    !clear <number> - Clears a specified number of messages from the channel.

    For any issues or additional help, feel free to ask!
    """

    await ctx.send(help_message)
```

更新內容說明：

1. 分組顯示：將命令分為不同的類別（General, Fun, Music, Game, Poll, User, Moderation），使其更易於閱讀和理解。
2. 新增命令描述：為每個命令添加了簡短的描述，讓用戶清楚每個命令的功能。



程式碼除錯

將程式碼出現錯誤傳送給chat on進行除錯，並依照結果進行修改

你遇到的錯誤信息表明 `message` 對象（應該是 `ctx`）沒有 `content` 屬性。這是因為在 Discord 的 `commands` 擴展中，命令處理通常使用 `ctx` 對象，而不是直接使用消息對象。

以下是修正後的 `play` 命令，使用 `ctx` 來獲取消息內容：

修正的 `play` 命令

Code

```
@bot.command()
async def play(ctx):
    queues = {}
    voice_clients = {}
    yt_dl_options = {"format": "bestaudio/best"}
    ytdl = yt_dlp.YoutubeDL(yt_dl_options)

    ffmpeg_options = {
        'before_options': '-reconnect 1 -reconnect_streamed 1 -reconnect_delay_max 5',
        'options': '-vn -filter:a "volume=0.25"'
    }

    # 確保使用者在語音頻道中
    if ctx.author.voice:
        try:
            voice_client = await ctx.author.voice.channel.connect()
            voice_clients[ctx.guild.id] = voice_client
        except Exception as e:
            print(f"Error connecting to voice channel: {e}")
            await ctx.send(f"Error connecting to voice channel: {e}")
            return

        try:
            # 從消息中獲取 URL
            url = ctx.message.content.split()[1] # 確保有提供 URL
            loop = asyncio.get_event_loop()
            data = await loop.run_in_executor(None, lambda: ytdl.extract_info(url,
                download=False))

            song = data['url']
            player = discord.FFmpegPCMAudio(song, **ffmpeg_options)

            voice_clients[ctx.guild.id].play(player)
            await ctx.send(f'Now playing: {data["title"]}')
        except Exception as e:
            print(f"Error playing audio: {e}")
            await ctx.send(f"Error playing audio: {e}")
```


你的程式碼大體上是正確的，但在播放音頻時可能會遇到一些潛在的問題，特別是在使用 `discord.FFmpegOpusAudio` 和 `FFmpegExtractAudio` 的時候。以下是一些建議和潛在改進的地方，以幫助你更好地播放音頻和處理可能的問題：

1. 使用 `FFmpegPCMAudio` 而不是 `FFmpegOpusAudio`

由於你已經將音頻轉換為 MP3 格式，使用 `FFmpegPCMAudio` 會更合適。如下所示：

Code



```
ctx.voice_client.play(discord.FFmpegPCMAudio(audio_file, **ffmpeg_options))
```

2. 檢查 `ffmpeg` 的安裝

確保你在執行機器人的環境中正確安裝了 `ffmpeg`。在 [Google Colab](#) 或其他環境中，你可以使用以下命令來安裝 `ffmpeg`：

Code



```
!apt-get install ffmpeg
```

3. 確保音頻文件存在

當你嘗試播放音頻時，確保音頻文件已經成功下載並且存在於指定位置。你可以在播放之前添加一個檢查：

Code



```
import os
```

```
if os.path.exists(audio_file):
    ctx.voice_client.play(discord.FFmpegPCMAudio(audio_file, **ffmpeg_options))
else:
    await ctx.send(f'Audio file {audio_file} does not exist.')
```

4. 監控錯誤

在播放音頻時，使用更詳細的錯誤處理來捕獲和顯示具體的錯誤信息。這樣可以幫助你更好地理解問題所在。

5. 完整的 `play` 命令示範

以下是更新後的 `play` 命令示範：

Code



```
@bot.command()
async def play(ctx, url):
    if ctx.author.voice:
        channel = ctx.author.voice.channel
        if ctx.voice_client is None:
            await channel.connect()

        with youtube_dl.YoutubeDL(ydl_opts) as ydl:
            try:
                info = ydl.extract_info(url, download=True) # 嘗試下載音頻
                audio_file = f"{info['title']}.mp3" # 使用音頻的標題作為文件名

                if os.path.exists(audio_file):
                    ctx.voice_client.play(discord.FFmpegPCMAudio(audio_file,
                                                                    **ffmpeg_options))
                    await ctx.send(f'Now playing: {info["title"]}')
                else:
                    await ctx.send(f'Audio file {audio_file} does not exist.')

            except Exception as e:
                await ctx.send(f'Error playing audio: {str(e)}')
    else:
        await ctx.send("You're not in a voice channel!")
```

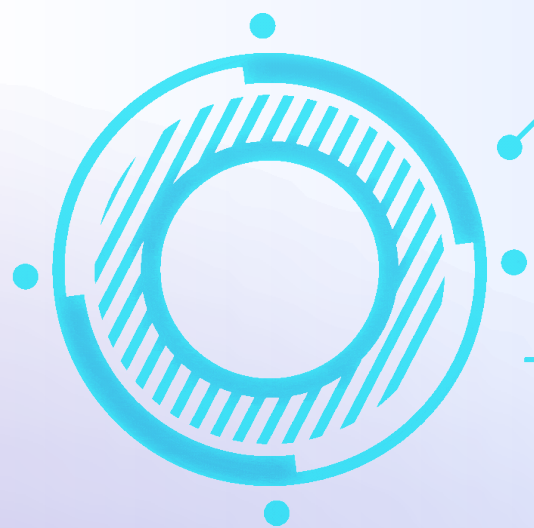



KEY

製作過程

by Colab





部分程式碼

```
import discord
from discord.ext import commands
import pyjokes
import yt_dlp
import nest_asyncio
import nacl
import asyncio
import random

nest_asyncio.apply()

intents = discord.Intents.default()
intents.members = True
intents.message_content = True
intents.voice_states = True

# 設定機器人前綴
bot = commands.Bot(command_prefix='!', intents = intents)

# 更新 ydl_opts
ydl_opts = {'format': 'bestaudio/best'}

ffmpeg_options = {'before_options' : '-reconnect 1 -reconnect_streamed 1 -reconnect_delay_max 5', 'options' : '-vn -filter:a'}

# 當機器人成功啟動時會執行的事件
@bot.event
async def on_ready():
    print(f'Logged in as {bot.user.name}')

@bot.command()
async def hello(ctx):
    await ctx.send('Hello! How can I help you today?')
```

```
@bot.command()
async def help_command(ctx):
    help_message = """
    Here are the commands you can use:

    **General Commands:**
    !hello - Greet you.
    !help_command - Shows this help message.

    **Fun Commands:**
    !joke - Tells a random joke.
    !rps <rock/paper/scissors> - Play rock, paper, scissors against the bot.

    **Music Commands:**
    !play <url> - Plays music from the given YouTube URL.
    !leave - Makes the bot leave the voice channel.

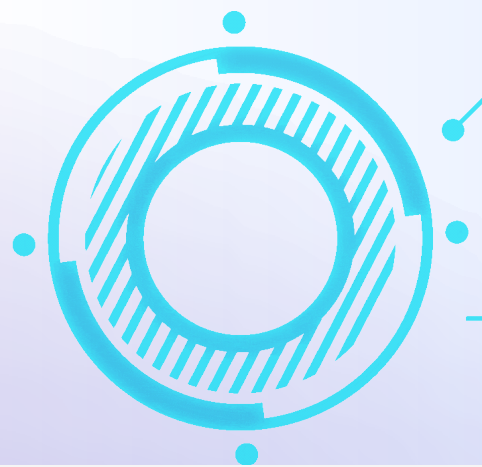
    **Game Commands:**
    !start_game - Starts a number guessing game (0-100).
    !end_game - Ends the current number guessing game.

    **Poll Commands:**
    !poll <question> <option1> <option2> [option3] ... - Creates a poll.

    **User Commands:**
    !userinfo [@user] - Displays information about a user.

    **Moderation Commands:**
    !clear <number> - Clears a specified number of messages from the channel.

    For any issues or additional help, feel free to ask!
    """
    await ctx.send(help_message)
```



遊戲程式碼

```
active_games = {}

@bot.command()
async def start_game(ctx):
    secret_number = random.randint(0, 100) # 隨機生成 0 到 100 的數字
    active_games[ctx.guild.id] = secret_number # 儲存正在進行的遊戲
    await ctx.send("A new game has started! Guess the number between 0 and 100.")

    def check(m):
        return m.author == ctx.author and m.channel == ctx.channel

    while ctx.guild.id in active_games: # 確保遊戲仍在進行中
        try:
            guess = await bot.wait_for('message', check=check, timeout=60) # 等待用戶的猜測
            guess_number = int(guess.content)

            if guess_number < 0 or guess_number > 100:
                await ctx.send("Please guess a number between 0 and 100.")
                continue

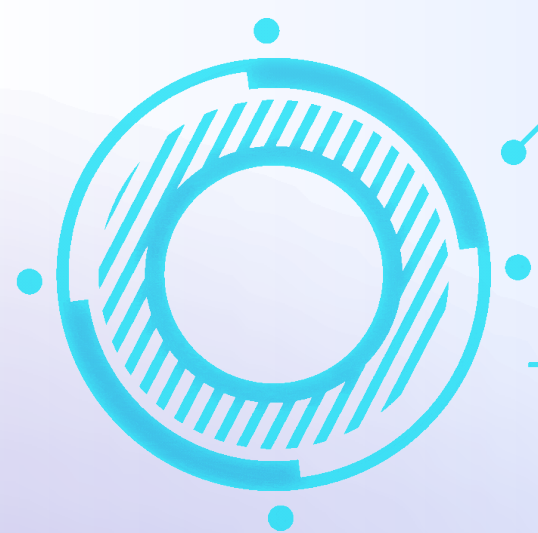
            if guess_number < secret_number:
                await ctx.send("Too low! Try again.")
            elif guess_number > secret_number:
                await ctx.send("Too high! Try again.")
            else:
                await ctx.send(f"Congratulations! You guessed the number: {secret_number}")
                del active_games[ctx.guild.id] # 刪除遊戲
                break

        except ValueError:
            if guess.content == '!end_game':
                pass
            else:
                await ctx.send("Please enter a valid number.")
        except asyncio.TimeoutError:
            await ctx.send(f"Time's up! The secret number was: {secret_number}")
            del active_games[ctx.guild.id] # 刪除遊戲
            break
```

```
@bot.command()
async def end_game(ctx):
    if ctx.guild.id in active_games:
        del active_games[ctx.guild.id] # 結束遊戲
        await ctx.send("The game has ended. You can start a new game with !start_game")
    else:
        await ctx.send("There is no active game to end.")

# 在這裡放置你的 Discord 機器人 token
# bot.run('YOUR_BOT_TOKEN')

@bot.command()
@commands.has_permissions(manage_messages=True)
async def clear(ctx, amount: int):
    await ctx.channel.purge(limit=amount + 1) # +1 to remove the command message
    await ctx.send(f"Cleared {amount} messages.", delete_after=5)
```



程式碼優化

調整猜數字遊戲功能，為每個使用者維護一個獨立的遊戲狀態，使其能夠同時進行獨立遊玩，並在結束時終止自己的遊戲。



怎麼讓多個使用者同時遊玩可以分別停止遊戲



要讓多個使用者在 Discord 中同時參加終極密碼遊戲並且能夠各自結束遊戲，你需要為每個使用者維護一個獨立的遊戲狀態。這可以通過使用一個字典來存儲每個使用者的遊戲狀態，而不是僅僅使用伺服器 ID。下面是對你的代碼進行修改以支持這一功能的示例：

修改後的代碼

Code

```
import random
import asyncio
from discord.ext import commands

bot = commands.Bot(command_prefix='!')

# 儲存正在進行的遊戲，使用者 ID 作為鍵
active_games = {}

@bot.command()
async def start_game(ctx):
    secret_number = random.randint(0, 100) # 隨機生成 0 到 100 的數字
    active_games[ctx.author.id] = secret_number # 儲存正在進行的遊戲
    await ctx.send(f"A new game has started for {ctx.author.mention}! Guess the number between 0 and 100.")

def check(m):
    return m.author == ctx.author and m.channel == ctx.channel

while ctx.author.id in active_games: # 確保遊戲仍在進行中
    try:
        guess = await bot.wait_for('message', check=check, timeout=60) # 等待用戶的猜測
        guess_number = int(guess.content)

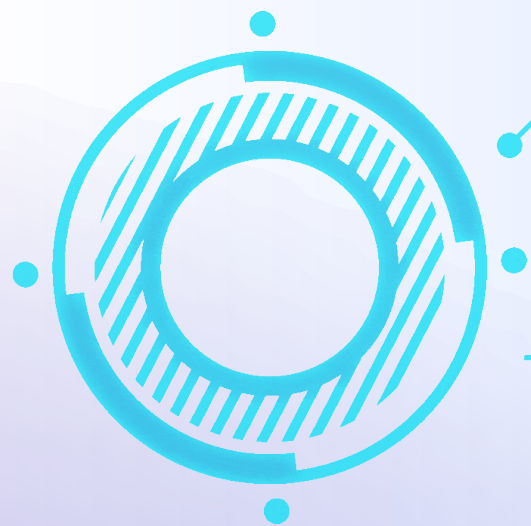
        if guess_number < 0 or guess_number > 100:
            await ctx.send("Please guess a number between 0 and 100.")
            continue

        if guess_number < secret_number:
            await ctx.send("Too low! Try again.")
        elif guess_number > secret_number:
            await ctx.send("Too high! Try again.")
```

修改說明：

1. 使用者 ID 作為鍵：將 `active_games` 的鍵更改為使用者的 ID (`ctx.author.id`)，這樣每個使用者的遊戲狀態是獨立的。
2. 獨立遊戲邏輯：在 `start_game` 和 `end_game` 命令中，通過使用者 ID 來檢查和管理遊戲狀態。
3. 結束遊戲：在猜測過程中，檢查用戶是否輸入了 `!end_game`，如果是則呼叫 `end_game` 函數來結束遊戲。

這樣，每個使用者都可以獨立地參加和終止自己的遊戲。希望這能滿足你的需求！如果有其他問題或需求，請隨時告訴我！



程式碼優化

```
active_games = {}

@bot.command()
async def start_game(ctx):
    secret_number = random.randint(0, 100) # 隨機生成 0 到 100 的數字
    active_games[ctx.author.id] = secret_number # 儲存正在進行的遊戲
    await ctx.send(f"A new game has started for {ctx.author.mention}! Guess the number between 0 and 100.")

    def check(m):
        return m.author == ctx.author and m.channel == ctx.channel

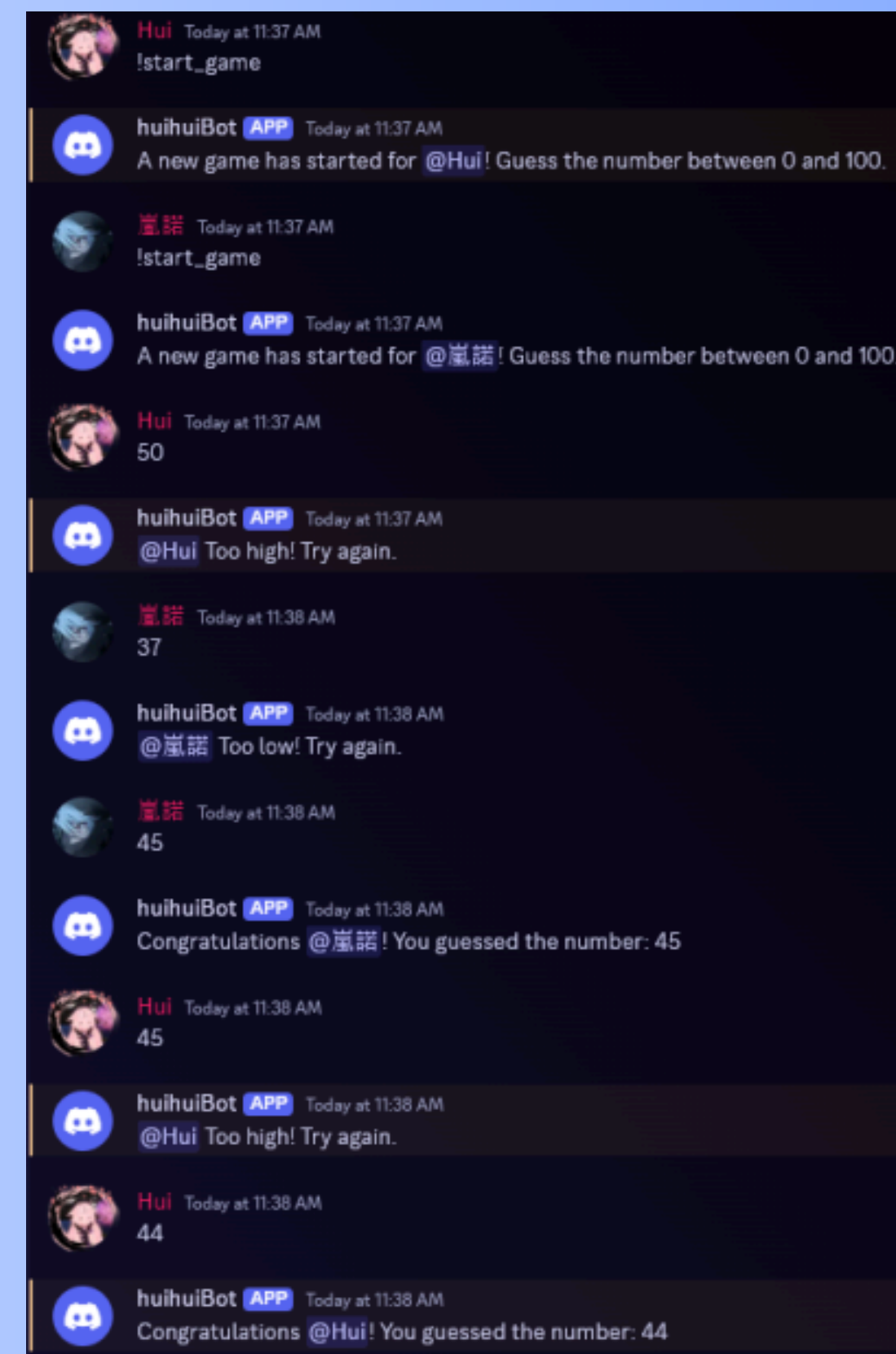
    while ctx.author.id in active_games: # 確保遊戲仍在進行中
        try:
            guess = await bot.wait_for('message', check=check, timeout=60) # 等待用戶的猜測
            guess_number = int(guess.content)

            if guess_number < 0 or guess_number > 100:
                await ctx.send("Please guess a number between 0 and 100.")
                continue

            if guess_number < secret_number:
                await ctx.send("Too low! Try again.")
            elif guess_number > secret_number:
                await ctx.send("Too high! Try again.")
            else:
                await ctx.send(f"Congratulations {ctx.author.mention}! You guessed the number: {secret_number}")
                del active_games[ctx.author.id] # 刪除遊戲
                break

        except ValueError:
            if guess.content == '!end_game':
                await end_game(ctx) # 呼叫結束遊戲函數
                break
            else:
                await ctx.send("Please enter a valid number.")
        except asyncio.TimeoutError:
            await ctx.send(f"Time's up! The secret number was: {secret_number}")
            del active_games[ctx.author.id] # 刪除遊戲
            break

@bot.command()
async def end_game(ctx):
    if ctx.author.id in active_games:
        del active_games[ctx.author.id] # 結束遊戲
        await ctx.send(f"The game has ended for {ctx.author.mention}. You can start a new game with !start_game.")
    else:
        await ctx.send("There is no active game to end for you.")
```



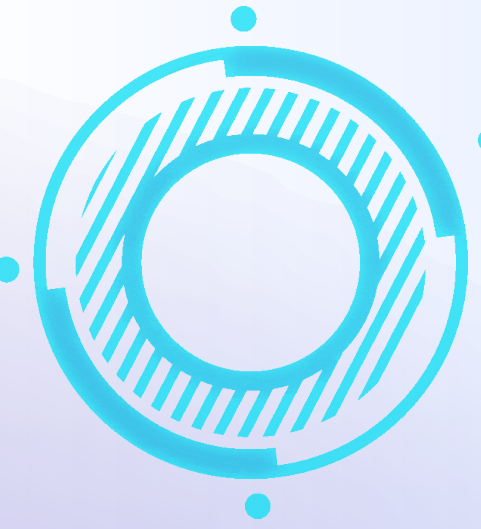


KEY

成品操作

Discord bot





指令表



Hui Today at 9:53 AM
!help_command



huihuiBot **APP** Today at 9:53 AM
Here are the commands you can use:

General Commands:

!hello - Greet you.
!help_command - Shows this help message.

Fun Commands:

!joke - Tells a random joke.
!rps <rock/paper/scissors> - Play rock, paper, scissors against the bot.

Music Commands:

!play <url> - Plays music from the given YouTube URL.
!leave - Makes the bot leave the voice channel.

Game Commands:

!start_game - Starts a number guessing game (0-100).
!end_game - Ends the current number guessing game.

Poll Commands:

!poll <question> <option1> <option2> [option3] ... - Creates a poll.

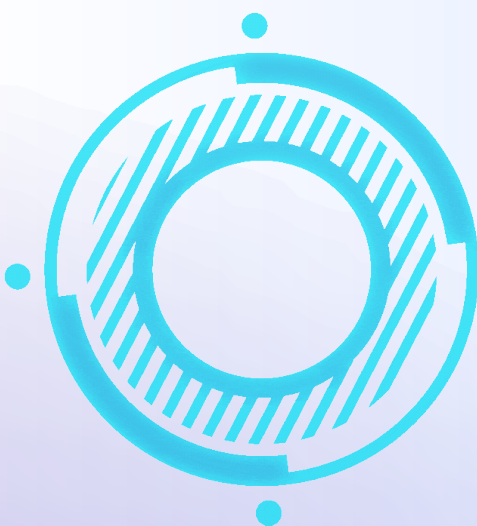
User Commands:

!userinfo [@user] - Displays information about a user.

Moderation Commands:

!clear <number> - Clears a specified number of messages from the channel.

For any issues or additional help, feel free to ask!



指令展示



Hui Today at 9:55 AM

!joke



huihuiBot **APP** Today at 9:55 AM

Why do Java programmers have to wear glasses? Because they don't see sharp.



Hui Today at 9:53 AM

!hello



huihuiBot **APP** Today at 9:53 AM

Hello! How can I help you today?



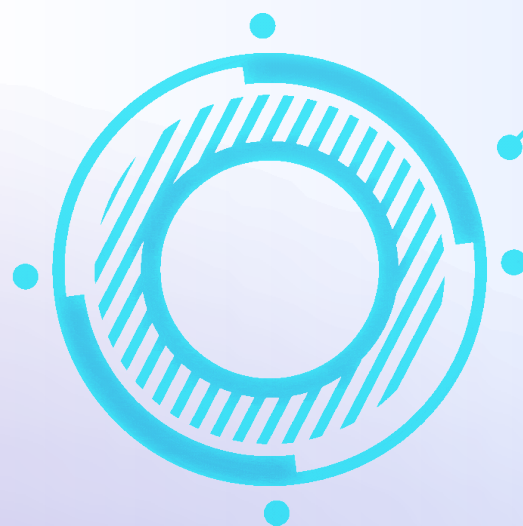
Hui Today at 9:56 AM

!rps rock



huihuiBot **APP** Today at 9:56 AM

You chose rock, I chose scissors. You win!



指令展示



Hui Today at 10:09 AM

!poll Lunch? Steak Pasta Hotpot

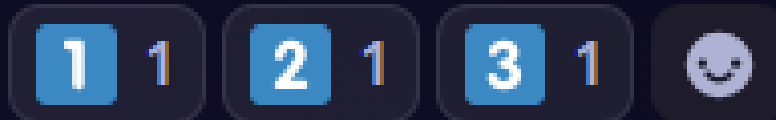


huihuiBot **APP** Today at 10:09 AM

Lunch?

1. Steak
2. Pasta
3. Hotpot

React with the corresponding number to vote!



Hui Today at 10:12 AM

!userinfo

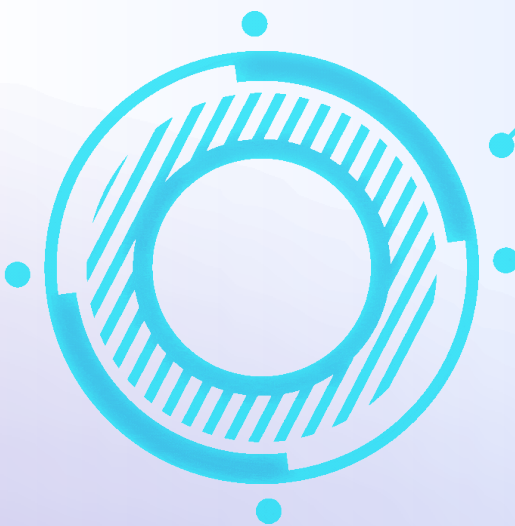


huihuiBot **APP** Today at 10:12 AM

huihuilaa's Info

ID	Joined at	Created at
1120689886341644308	2024-10-18 18:11:05	2023-06-20 12:21:30
Roles	Status	Nickname
Admin	Offline	Hui





音樂播放

```
2024-12-13 01:47:34 INFO discord.voice_state The voice handshake is being terminated f
INFO:discord.voice_state:The voice handshake is being terminated for Channel ID 12968983657239
2024-12-13 01:47:34 INFO discord.player ffmpeg process 2352 has not terminated. Waitin
INFO:discord.player:ffmpeg process 2352 has not terminated. Waiting to terminate...
2024-12-13 01:47:34 INFO discord.player ffmpeg process 2352 should have terminated wit
INFO:discord.player:ffmpeg process 2352 should have terminated with a return code of -9.
2024-12-13 01:48:04 INFO discord.voice_state Connecting to voice...
INFO:discord.voice_state:Connecting to voice...
2024-12-13 01:48:04 INFO discord.voice_state Starting voice handshake... (connection a
INFO:discord.voice_state:Starting voice handshake... (connection attempt 1)
2024-12-13 01:48:05 INFO discord.voice_state Voice handshake complete. Endpoint found:
INFO:discord.voice_state:Voice handshake complete. Endpoint found: hongkong11011.discord.media
2024-12-13 01:48:06 INFO discord.voice_state Voice connection complete.
INFO:discord.voice_state:Voice connection complete.
[youtube] Extracting URL: https://www.youtube.com/watch?v=rfTgO9rpqck
[youtube] rfTgO9rpqck: Downloading webpage
[youtube] rfTgO9rpqck: Downloading ios player API JSON
[youtube] rfTgO9rpqck: Downloading mweb player API JSON
[youtube] rfTgO9rpqck: Downloading m3u8 information
```

 **Hui** Today at 9:48 AM
!play <https://www.youtube.com/watch?v=rfTgO9rpqck>

YouTube
LatinHype
Glass Animals - Heat Waves

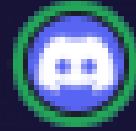


 **huihuiBot** **APP** Today at 9:48 AM
Now playing: Glass Animals - Heat Waves

 **Test**   

Set a channel status 

 **Hui** 

 **huihuiBot**

心得與未來展望

經過這次機器人實作，讓我們發現雖然AI可以解決大部分問題，但依然還是需要人工親自檢查並處理剩餘錯誤，如製作過程中，程式碼雖然已經交由Chat gpt進行除錯，實際進行時依然會出現Bug，這時就需要親自進行Debug，且正確下達指令讓Chat gpt能正確執行出想要的成果也是在製作中的一大難題，往往需要經過多次嘗試、換句話說才能使AI達成自己真正想要的目的，希望未來能增加更多功能。





THANK YOU!

